10. The length of the tangent from (4,5) to the circle $x^2 + y^2 = 16$ is

(a) 4 (b) 5 (c) 16 (d) 25

11. The focus of the parabola $x^2 = 16y$ is

(a) (4 ,0) (b) (–4 , 0)

(c) (0, 4) (d) (0, –4)

12. Length of the latus rectum of the parabola $y^2 = -25x$.

(a) 25 (b) –5 (c) 5 (d) –25

13. The centre of the circle $x^2 + y^2 - 2x + 2y - 9 = 0$ is

(a) (1 ,1) (b) (–1, –1)

(c) (–1,1) (d) (1, –1)

14. The equation of the circle with centre on the x axis and passing through the origin is

(a) $x^2 - 2ax + y^2 = 0$

(b) $y^2 - 2ay + x^2 = 0$

(c) $x^2 + y^2 = a^2$

(d) $x^2 - 2ay + y^2 = 0$

15. If the centre of the circle is $(-a, -b)$ and radius is $\sqrt{a^2 - b^2}$, then the equation of circle is

(a) $x^2 + y^2 + 2ax + 2by + 2b^2 = 0$

(b) $x^2 + y^2 + 2ax + 2by - 2b^2 = 0$

(c) $x^2 + y^2 - 2ax - 2by - 2b^2 = 0$

(d) $x^2 + y^2 - 2ax - 2by + 2b^2 = 0$

16. Combined equation of co-ordinate axes is

(a) $x^2 - y^2 = 0$ (b) $x^2 + y^2 = 0$

(c) $xy = c$ (d) $xy = 0$

17. $ax^2 + 4xy + 2y^2 = 0$ represents a pair of parallel lines then 'a' is

(a) 2 (b) –2 (c) 4 (d) –4

18. In the equation of the circle $x^2 + y^2 = 16$ then y intercept is (are)

(a) 4 (b)16 (c) ±4 (d) ±16

19. If the perimeter of the circle is 8π units and centre is (2,2) then the equation of the circle is

(a) $(x-2)^2 + (y-2)^2 = 4$

(b) $(x-2)^2 + (y-2)^2 = 16$

(c) $(x-4)^2 + (y-4)^2 = 2$

(d) $x^2 + y^2 = 4$

20. The equation of the circle with centre (3, –4) and touches the x – axis is

(a) $(x-3)^2 + (y-4)^2 = 4$

(b) $(x-3)^2 + (y+4)^2 = 16$

(c) $(x-3)^2 + (y-4)^2 = 16$

(d) $x^2 + y^2 = 16$

21. If the circle touches x axis, y axis and the line $x = 6$ then the length of the diameter of the circle is

(a) 6 (b)3 (c) 12 (d) 4

22. The eccentricity of the parabola is

(a) 3 (b) 2 (c) 0 (d) 1

23. The double ordinate passing through the focus is

(a) focal chord (b) latus rectum
(c) directrix (d) axis

24. The distance between directrix and focus of a parabola $y^2 = 4ax$ is

(a) a (b) $2a$ (c) $4a$ (d) $3a$

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